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Variable Recurrent Unit

Architecture Diagram

φ = 4/π ≈ 1.2732 · α = π/4 ≈ 0.7854 · D_CAP = 4 · n_dims = 5

φ-scaled (memory field)
π-scaled (input field)
anchor (learnable)
ω gate
C field (carry state)
data flow

D P P U C E L L n_dims = 5 · dim_size = hidden // 5 TOKEN x_t ∈ vocab Emb W_x → x_proj h_prev hidden state C_prev carry field compute_delta |h_slice| / σ(h_slice) δᵢ = tanh(mag / spread) φ_dyn(δ) φ·e^−δ + (1−e^−δ) memory field π_dyn(δ) 4 − (4−π)·e^−δ input field ω(δ) = σ(φ_dyn · π_dyn / (1 + δ)) gate — spectral product controls oscillation tightness sigmoid(ω) ∈ (0, 1) h_rec (φ_dyn / φ) · W_h[i](h) recurrent weight path x_rec (π_dyn / π) · x_proj[slice] input projection path anchor C[:,i] · anchor_ref learnable · init: φ free → ~1.27–1.31 ⊕ tanh(h_rec + x_rec + anchor) · ω(δ) → h_i (single dimension slice output) cat(h_0…h_4) W_out → h_new C_new update clamp(tanh(0.1C + 0.01W_c(h)), −0.5, 0.5) h_new → h_prev next step W_out → logits vocab size recurrent [i = 0..4, each dim_size = hidden/5] [slice]

φ Memory Field

φ = 4/π ≈ 1.2732

Scales the recurrent weight path. Sits at the oscillatory boundary — values persist without decaying or exploding. The attractor that defines the architecture’s identity. Self-organises to ~1.27–1.31 on arithmetic tasks when unconstrained.

π Input Field

α = π/4 ≈ 0.7854 (reciprocal dual)

Scales the input projection path. Stays in the contraction region — decays between events, spikes when carry fires. $\varphi \times \alpha = 1.0$ by construction. These are mathematical reciprocal duals, not arbitrary constants.

Learnable Anchor

`anchor = C[:,i] · anchor_ref`

A learnable scalar parameter, one per cell. Initialised at φ or 0.420610. Left free (no regularisation), it self-organises to the geometry the task requires. Acts as a grounding term that prevents mode collapse in the recurrent field.

ω Gate

`ω(δ) = φ_dyn · π_dyn / (1 + δ)`

The spectral product controls oscillation tightness. Passed through sigmoid to produce a soft gate $\in (0,1)$. δ is derived from the local hidden state slice — so the gate is self-referential. Not an external control signal.

C Field (Carry State)

`clamp(tanh(0.1C + 0.01W_c(h)), -0.5, 0.5)`

Separate from h. Clamped to [−0.5, 0.5] to prevent field explosion. Feeds the anchor. Must occupy separate recurrent space from h — sharing causes carry events to contaminate W_h's eigenstructure.

Dual-Field Geometry

`n_dims = D_CAP + 1 = 5 slots`

The hidden state h is partitioned into 5 slices of dim_size = hidden//5. Each slice has its own delta, its own phi_dyn/pi_dyn evaluation, and its own contribution to h_new. In v23 (next): explicit register slots for arithmetic operands within these fields.
